



Ouail Zakary

Postdoctoral Researcher, Ph.D. in Physics
NMR Research Unit, Faculty of Science, University of Oulu
P.O. Box 8000
FIN-90014 University of Oulu
Finland
E-mail: ouail.zakary@oulu.fi
Personal Website: <https://ozakary.github.io>
Website: [Ouail Zakary](#)
GitHub: [ozakary](#)
Scopus ID: [58662986700](#)
ResearcherID: [OSI-7733-2025](#)
ORCID: [0000-0002-7793-3306](#)
Google Scholar: [Ouail Zakary](#)

List of Publications

List date: August 24, 2026

Note: Corresponding authorship is marked with (*)

A. Peer-reviewed scientific articles

A.1. Published articles

1. **Zakary, O.***; Body, M.*; Charpentier, T.; Sarou-Kanian, V.; Legein, C. Structural Modeling of O/F Correlated Disorder in TaOF₃ and NbOF_{3-x}(OH)_x by Coupling Solid-State NMR and DFT Calculations. *Inorg. Chem.* **2023**, *62*, 16627–16640.
[DOI: [10.1021/acs.inorgchem.3c02844](https://doi.org/10.1021/acs.inorgchem.3c02844)] [[Supporting Data](#)]
2. **Zakary, O.***; Body, M.; Sarou-Kanian, V.; Arnaud, B.; Corbel, G.*; Legein, C. Different Magnitudes of Second-Order Jahn-Teller Effect in Isostructural NaMO₂F₂ (M= Nb, Ta) Oxyfluorides. *J. Alloys Compd.* **2025**, *1010*, 177457.
[DOI: [10.1016/j.jallcom.2024.177457](https://doi.org/10.1016/j.jallcom.2024.177457)] [[Supporting Data](#)] [[Supporting Code](#)]
3. **Zakary, O.***; Body, M.; Charpentier, T.; Sarou-Kanian, V.; Legein, C. Revealed Preferential Short-Range Anion Ordering in Disordered RbM₂O₅F (M= Nb, Ta) Pyrochlore-Type Oxyfluorides. *Inorg. Chem.* **2025**, *64*, 5764–5777.
[DOI: [10.1021/acs.inorgchem.5c00615](https://doi.org/10.1021/acs.inorgchem.5c00615)] [[Supporting Data](#)] [[Supporting Code](#)]
4. **Zakary, O.***; Lantto, P. Equivariant Neural Networks Reveal How Host–Guest Interactions Shape ¹²⁹Xe NMR in Porous Liquids. *J. Phys. Chem. Lett.* **2025**, *16*, 12095–12103. (Selected as *Journal Supplementary Cover*)
[DOI: [10.1021/acs.jpcclett.5c02846](https://doi.org/10.1021/acs.jpcclett.5c02846)] [[Supporting Data](#)] [[Supporting Code](#)] [[Journal Cover](#)]
5. Laurila, O.; Jacklin, T.; **Zakary, O.***; Lantto, P.* Machine Learning-Accelerated Path Integral Molecular Dynamics and ¹³C NMR Simulations Unlock New Insights Into Quantum Effects in C₆₀ Fullerene. *J. Phys. Chem. A* **2026**, *130*, 2169–2181.
[DOI: [10.1021/acs.jpca.6c00238](https://doi.org/10.1021/acs.jpca.6c00238)] [[Supporting Data](#)] [[Supporting Code](#)]

A.2. Pre-prints (submitted articles)

6. **Zakary, O.***; Yin, W.; Aryal, N.* Local Symmetry Breaking and Correlated Fluctuations in Quantum Materials from Atomistic Foundation Models. *ChemRxiv* **2026**, 0330, *Submitted for publication*.
[DOI: [10.26434/chemrxiv.15001387/v2](https://doi.org/10.26434/chemrxiv.15001387/v2)] [[Supporting Data](#)] [[Supporting Code](#)]

A.3. Articles in preparation

7. **Zakary, O.***; Jacklin, T.; Lantto, P. Transport Properties of Carbon Nanotubes Unlocked by Machine Learning-Accelerated Molecular Dynamics and Noble Gas NMR Simulations. 2026, *In preparation*.
8. **Zakary, O.***; Body, M.; Sarou-Kanian, V.; Legein, C. Hidden Short-Range Order in Mixed-Anion Sodium-Oxyfluorotantalates. 2026, *In preparation*.
9. Laurila, O.*; **Zakary, O.***; Lantto, P.* Machine Learning for Computing NMR Observables Sensitive to Quantum Effects. 2026, *In preparation*.
10. Laurila, O.*; **Zakary, O.***; Lantto, P.* ^3He NMR Monomer-Dimer Shift in $\text{He}_m@\text{C}_{60}^{n-}$ and $\text{He}_m@\text{C}_{70}^{n-}$ ($m = 1, 2$, and $n = 0, 6$) Endohedral Fullerenes from Machine Learning-Guided Simulations. 2026, *In preparation*.
11. **Zakary, O.***; Mailhiet, S.E.; Selent, A.; Mareš, J.; Lantto, P.*; Telkki, V.-V.* Unifying the Description of Type-II Porous Liquids Through Machine Learning-Accelerated Molecular Dynamics and ^{129}Xe NMR: Simulations Informing Experiments. 2026, *In preparation*.
12. Hintsanen, M.; Lantto, P.; **Zakary, O.*** Insights into the Density Maximum of Water from Machine Learned ^{129}Xe NMR and Molecular Dynamics. 2027, *In preparation*.
13. **Zakary, O.*** A Machine Learning Workflow for Solving Disorder in Heteroanionic Materials. 2027, *In preparation*.
14. **Zakary, O.*** A Foundation Model for NMR Parameters in Materials Chemistry. 2027, *In preparation*.
15. Laurila, O.*; **Zakary, O.***; Lantto, P.* Machine Learning-Driven Path Integral Molecular Dynamics and ^{13}C NMR Simulations Enable the Assignments of Singlet NMR in the Case of High Molecular Symmetry. 2027, *In preparation*.
16. Selent, A.; **Zakary, O.***; Mareš, J.; Lantto, P.*; Telkki, V.-V.* Probing R3@DMSO and R3S@H₂O Porous Liquids Combining ^{129}Xe NMR and Machine Learning-Accelerated Simulations. 2027, *In preparation*.
17. Chennabassapa, M.; Squires, A.G.; Demourgues, A.; Penin, N.; Durand, E.; Li, W.; Legein, C.; **Zakary, O.**; Body, M.; Charpentier, T.; Borkiewicz, O.J.; Morgan, B.J.; Scanlon, D.O.; Dambournet, D.* Hexagonal-Tungsten-Bronze TiOF_2 : Synthesis, Crystal Structure and Anionic Short-Range Ordering. 2027, *In preparation*.

B. Non-refereed scientific articles

B.1. Conference proceedings and presentations

B.1.1. Contributed talks

1. **Zakary, O.**; Lantto, P. Efficient and Accurate Local Equivariant Deep Neural Network Interatomic Potential for Large-Scale Porous Liquids Simulations. *Winter School in Theoretical Chemistry*, 09-12 Dec. **2024**, Helsinki, Finland.
[DOI: [10.6084/m9.figshare.29757140.v1](https://doi.org/10.6084/m9.figshare.29757140.v1)]
2. **Zakary, O.**; Lantto, P. Equivariant Neural Networks Reveal How Host–Guest Interactions Shape Xenon NMR Chemical Shift in Porous Organic Cages. *The 21th European Magnetic Resonance Congress (EUROMAR)*, 06-10 July **2025**, Oulu, Finland.
[DOI: [10.6084/m9.figshare.29757203.v1](https://doi.org/10.6084/m9.figshare.29757203.v1)]
3. **Zakary, O.** Understanding Host-Guest Interactions in Type-II Porous Liquids using Machine Learning-Accelerated Molecular and NMR Parameters Simulations. *Computational Chemistry Days (CCD)*, 25-26 May **2026**, Helsinki, Finland.
[DOI: [10.6084/m9.figshare.32519391](https://doi.org/10.6084/m9.figshare.32519391)]

B.1.2. Other talks

4. **Zakary, O.**; Body, M.; Charpentier, T.; Legein, C. Structural Modeling of Disordered Inorganic Oxyfluorides by Coupling Solid State NMR and DFT Calculations. *Réunion RMN Grand Bassin Parisien*, 19 Nov. **2021**, Orsay, France.
[DOI: [10.6084/m9.figshare.27111955.v2](https://doi.org/10.6084/m9.figshare.27111955.v2)]
5. **Zakary, O.**; Body, M.; Charpentier, T.; Sarou-Kanian, V.; Legein, C. Structural Modeling of O/F Correlated Disorder in NbOF₃ and TaOF₃. *Alpine Conference on Magnetic Resonance on Solids*, 04-08 Sept. **2022**, Chamonix-Mont-Blanc, France.
[DOI: [10.6084/m9.figshare.27111496.v2](https://doi.org/10.6084/m9.figshare.27111496.v2)]
6. **Zakary, O.** Towards a Complete Structural Description of Disordered Inorganic Fluorides: The Crucial Role of Solid State NMR and DFT Calculations. *JED – 3MG, Journée de l'Ecole Doctorale*, 25 May **2023**, Nantes, France.
[DOI: [10.6084/m9.figshare.27115360.v2](https://doi.org/10.6084/m9.figshare.27115360.v2)]
7. **Zakary, O.**; Lantto, P. Modeling Porous Liquids with Machine-Learning-Assisted Molecular Dynamics. *Computational Chemistry Days (CCD)*, 27-28 May **2024**, Jyväskylä, Finland.
[DOI: [10.6084/m9.figshare.27105913.v1](https://doi.org/10.6084/m9.figshare.27105913.v1)]
8. **Zakary, O.**; Lantto, P. Scalable, Accurate, and Data-Efficient Machine Learning Potential models for Realistic Simulations of Porous Materials. *Chemical Solutions for Biological Challenges (CSBC)*, 10-12 June **2024**, Turku, Finland.
[DOI: [10.6084/m9.figshare.27105937.v2](https://doi.org/10.6084/m9.figshare.27105937.v2)]
9. **Zakary, O.**; Lantto, P. Efficient and Accurate Local Equivariant Deep Neural Network Interatomic Potential for Large-Scale Porous Liquids Simulations. *Magnetic Resonance in Porous Media 16 (MRPM)*, 26-30 Aug. **2024**, Tromsø, Norway.
[DOI: [10.6084/m9.figshare.27106033.v2](https://doi.org/10.6084/m9.figshare.27106033.v2)]

B.1.3. Poster presentations

10. **Zakary, O.**; Body, M.; Legein, C. Structural Modeling of Disordered Inorganic Fluorides. *JED – 3M445, Journée de l'Ecole Doctorale*, 27-28 May **2021**, Le Mans, France.
[DOI: [10.6084/m9.figshare.27115426.v1](https://doi.org/10.6084/m9.figshare.27115426.v1)]
11. **Zakary, O.**; Lantto, P. Combining Molecular Dynamics with Deep Neural Network Architectures for Realistic Simulations of Porous Liquids. *The 20th European Magnetic Resonance Congress (EUROMAR)*, 30 June - 04 July **2024**, Bilbao, Spain.
[DOI: [10.6084/m9.figshare.27105562.v2](https://doi.org/10.6084/m9.figshare.27105562.v2)]
12. **Zakary, O.**; Lantto, P. Machine Learning Driven Approach for Modeling Host-Guest Dynamics in Xenon-Based Porous Liquids. *Physics Days*, 26-28 Mar. **2025**, Oulu, Finland.
[DOI: [10.6084/m9.figshare.28684814.v1](https://doi.org/10.6084/m9.figshare.28684814.v1)]
13. **Zakary, O.**; Lantto, P. Local E(3)-Equivariant Neural Network Force Field for Modeling Host-Guest Interactions in Xenon-Based Porous Organic Cages. *Computational Chemistry Days*, 02-03 June **2025**, Espoo, Finland.
[DOI: [10.6084/m9.figshare.28684814.v1](https://doi.org/10.6084/m9.figshare.28684814.v1)]
14. **Zakary, O.**; Lantto, P. Host-Guest Dynamics in Porous Liquids Modeled Using E(3)-Equivariant Neural Networks. *The 13th Triennial Congress of the World Association of Theoretical and Computational Chemists (WATOC 2025)*, 21-27 June **2025**, Oslo, Norway.
[DOI: [10.6084/m9.figshare.28684814.v1](https://doi.org/10.6084/m9.figshare.28684814.v1)]
15. **Zakary, O.**; Lantto, P. Machine Learning-Accelerated Molecular Dynamics and NMR Parameter Simulations in Porous Materials. *The 22st European Magnetic Resonance Congress (EUROMAR)*, 28 June - 02 July **2026**, Gothenburg, Sweden.
[DOI: [10.6084/m9.figshare.32895569](https://doi.org/10.6084/m9.figshare.32895569)]

B.1.4. Schools

16. Moving ions with VASP. *University of Vienna*, 27-30 Sept. 2022, Online (Zoom)
17. Machine Learning of First Principles Observables. *Centre Européen de Calcul Atomique et Moléculaire (CECAM)*, 08-12 July 2024, Berlin, Germany.
18. Electronic Structure Theory. *Winter School in Theoretical Chemistry*, 15-18 Dec. 2025, Helsinki, Finland.

C. Theses

C.1. Doctoral dissertation

- **Zakary, O.** Structural Modeling of Oxygen-Fluorine Ordering in Transition Metal Inorganic Oxyfluorides. *Le Mans Université* **2023**.
[Thesis] [[Ph.D. Defense Details](#)]